Revised Scientific Report and Five-Hour Action Plan (v2.0)

Trastuzumab scFv Optimization Against HER2: 50 Proposed Candidates, Actual ipSAE, 24 GB GPU

Revised: 31 October 2026 — organizer parent sequence verified programmatically; earlier erroneous 244-aa claim withdrawn.
Objective: Generate **50 distinct valid scFv sequences as a computational proposal pool**; aim to provide **genuine ipSAE for all 50** if throughput permits; submit **only those candidates with actually computed ipSAE**, prioritizing **at least the final 15**. Include the design-method narrative and evidence provenance.
Compute constraint: One CUDA GPU with 24 GB of VRAM; **300 minutes total wall-clock**, including quality control and submission packaging.
Status: Research-backed **execution proposal**. The 50 sequence slots in the accompanying CSV are empty placeholders; no candidate sequence, Boltz structure, ipSAE value, or laboratory result has been generated by this report. All computational budgets depend on a local parent microbenchmark.

Executive summary: recommended decision

Use the existing keyshavmor/Catalyst2026Hackathon AbLang2 → ESM2-3B → Boltz-2 infrastructure, but **do not run the previous production orchestrator unchanged**. The existing code is hardwired in multiple places for Cetuximab–EGFR, 241-residue input, PDB 1YY9, and a 1,000,000-design generation funnel. The new experiment is Trastuzumab–HER2, with the **exact 242-residue organizer sequence supplied in chat**, PDB 1N8Z, **at least five mutations**, V_H–$(GGGGS)_3$–V_L, length **at most 260 amino acids**, and **15 lab-testing slots per team** (per the earlier challenge slide). The newer organizer instructions allow submitting as many proposed sequences as desired **but request ipSAE for each submitted design**. Thus 50 is an internal proposal target, *not* proof that 50 ipSAE-complete submissions can fit in 300 minutes.

The key innovation is **target-specific, evidence-weighted proposal generation** using known trastuzumab–HER2 affinity-variant literature, together with **parent-relative structural evaluation**, **liability controls**, and **diversity-aware final allocation**. Restrict initial engineering to fixed-length amino-acid substitutions (242 residues) for robustness. Build optional extensions only if the core pipeline is already working and the clock permits.

Recommended strategy: 1. Verify the **exact organizer-supplied parent**; map its variable chains to crystallographic chain B (heavy), chain A (light), and HER2 chain C in 1N8Z. Never assume the crystal Fab sequence and the contest scFv are identical. 2. Build a residue-level map (full scFv index ↔ V_H/V_L ↔ antibody numbering ↔ PDB coordinates ↔ epitope contact class), with explicit handling of unresolved residues and numbering schemes. 3. Generate 5–8-substitution variants through an AbLang2-guided local search, augmented by literature-informed CDR-H3 proposals and structurally conservative framework choices. 4. Produce a **50-row candidate manifest and FASTA early**, but mark **every ipSAE as missing until computed**; reserve 15 top nominations. Only export genuinely ipSAE-scored candidates to the scored-submission file. 5. ESM2-3B screen only a bounded, diverse shortlist; benchmark actual time instead of assuming 5,000 sequences are cheap. 6. **Use HER2 domain IV only** for all Boltz-2 cofolds, with parent and candidates under the same conditions; benchmark the parent first, then run **as many of the 50–100 shortlisted variants as fit**. Minimum practical priority is the nominated 15; never report an uncomputed score as measured. 7. Final ranking is **multi-objective**

and diversity-aware, not pure ipSAE maximization or a predicted K_D. Supply actual ipSAE + method narrative for every submitted candidate, and a separately labeled internal 50-row design pool.

CRITICAL: confirmed organizer rules and the corrected parent (v2 changes)

This section supersedes any earlier draft of the report. The user's supplied CSV is the immutable starting sequence. An earlier response incorrectly counted 244 aa; **the correct count is 242 aa**. Independently verified decomposition: **V_H 1–120 (120 aa), fixed linker 121–135 (15 aa), V_L 136–242 (107 aa)**. The four native cysteines occur at **full-scFv positions 22, 96, 158 and 223**, all 1-based. Verify these invariants again inside the execution environment; do not trust a copied summary.

Verbatim parent sequence (unbroken; the same string is in organizer_parent_scfv.fasta):

EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARIYPTNGYTRYADSVKG
RFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGDGFYAMDYWGQGTLVTVSSGGGGSGGGGSG
GGGSDIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSASFLYSGVPSRFSG
SRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGTKVEIK

SHA-256 of the 242-character sequence, without a newline:
0b9248c76a2b399791c47dbfa0d252b18284265f1dbca9ae8e66c0dd018c9f64. The challenge uses a single-chain **V_H–(GGGGS)$_3$–V_L** sequence and permits **up to 260 aa**. Our **self-imposed** quick-run policy is 242-aa, substitution-only sequences with **5–8 substitutions**. No linker edits. No edits to native cysteines at 22/96/158/223, and no introduced cysteines.

Mandatory organizer deliverables: (1) proposed sequences, each with >=5 mutations from the exact parent; (2) an actual **ipSAE score of each designed sequence submitted**; and (3) a clear scientific design-method explanation. The organizer's later text permits proposing many sequences and specifically discourages de novo approaches and excessive cofolding. The older slide says **15 candidates per team** will be lab-tested; these are compatible if more may be proposed than tested.

Prohibited or explicitly out-of-scope: BindCraft, RFdiffusion, BoltzGen and other de novo binder-generation tools; cofolding against the full ~630-aa HER2 ectodomain; mutating the linker or conserved cysteines; ranking solely by ipSAE; spending the full event on environment setup. **Allowed:** Boltz-2 for predicting the structure of an existing edited antibody sequence. **Cofold HER2 domain IV**, not the whole ectodomain. Domain-IV boundaries must be established by mapping PDB 1N8Z to UniProt P04626: do not borrow EGFR domain III from the prior repository. Preserve domain-IV epitope residues and structural context; treat engineered terminal artifacts explicitly.

Deliverable distinction (non-negotiable):

- internal_50_candidates.*: 50 valid proposals, including candidates whose ipSAE is still NOT_EVALUATED; **not yet a compliant fully-scored organizer submission**.
- submission_scored.*: **only** candidates with valid Boltz-2 structure + PAE outputs and an independently computed ipSAE, plus a narrative. If 50 are successfully scored, it may contain 50. If only 15 are scored, submit those 15 rather than attaching fictitious scores for 50. If fewer than 15 can be scored, declare the shortfall accurately and seek organizer help (including any offered Modal help).
- top_15_lab_nominations.*: a ranked subset of scored candidates. If a few shortlisted variants fail QC, replace them from the internal 50 and score replacements before submission.

ipSAE implementation: Boltz-2 does not ordinarily emit the DunbrackLab ipSAE as a standard field. Save pae_*.npz (full PAE) and the matching .cif / .pdb, then compute ipSAE with the **DunbrackLab/IPSAE** script or a verified wrapper. Record script commit/version, numerical PAE and distance cutoffs, direction/aggregation of interface scores (ipsae_min versus directional measures), chain order, sequence mapping and model seed. Do not relabel raw ipTM, iPAE, or a proxy as official ipSAE. Verify the tool's support for the actual Boltz output version with a parent test. See [R16].

1. Problem definition and scientific caveats

1.1 Challenge contract (from supplied slide)

Item	Contract / value	Interpretation
Target	Human HER2 extracellular domain; UniProt P04626, residues Thr23–Thr652 (630 residues)	Target sequence and numbering must be checked against UniProt and PDB.
Reference	PDB 1N8Z, HER2 ectodomain + trastuzumab Fab, ~2.52 Å	Experimental binding geometry; not the scFv model itself.
Parent	**Exactly 242 aa**, V_H 120 + linker 15 + V_L 107	Use supplied CSV verbatim; linker positions 121–135; cysteines 22, 96, 158, 223.
Rules	≥5 substitutions vs supplied parent, V_H–$(GGGGS)_3$–V_L, ≤260 aa; fixed linker and cysteines	Fast path: 5–8 substitutions, exactly 242 aa.
Experimental parent	K_D = **0.82 pM** by BLI, per slide	Very tight; robust kinetic fitting may be difficult.
Lab budget	15 variants/team; 45 for winner	Internal 50 is a shortlist, **not** 50 guaranteed lab tests.
Mandatory submission	Sequence(s), genuine ipSAE **for each submitted design**, design-method walkthrough	50 internal designs alone do not satisfy ipSAE deliverable.

For receptor–antibody interactions, the equilibrium dissociation constant is K_D = k_off/k_on. A design with higher model confidence need not have lower K_D. Prediction models largely do not provide calibrated scFv–HER2 dissociation constants at sub-picomolar scale. Treat 0.82 pM as the **contest-provided measured baseline**, not interchangeable with other publications' K_D or EC50 measured under different platforms, immobilization, formats, or assay conditions.

1.2 What external trastuzumab evidence supports

- **AlphaBind, 2025:** DMS-informed Trastuzumab-scFv optimization yielded a strongest CDR-H3-only candidate with an independently confirmed **2.8-fold KinExA affinity improvement**, from **7.43 to 2.74 pM**. The wider CDRH2–framework–H3 editing window did not outperform its parent in that particular comparison. This is strong evidence for **focused CDR-H3 engineering**, not proof that the same variants beat the challenge's **0.82 pM BLI** baseline [R6].
- **Mason et al., 2021:** Trastuzumab deep mutational libraries generated substantial labeled HER2-binding data and demonstrated supervised sequence screening [R7]. Such labels are often **binder/nonbinder** or enrichment, not same-platform sub-pM K_D. Avoid conflating them.

- **Qiu et al., 26 October 2026 (AbOptimizer):** In three of five experimentally tested trastuzumab CDR-H3 indel-bearing variants, antibody binding potency improved by an EC50-based assay; the best variant's apparent potency was ~3.2-fold better. **EC50 ≠ K_D**, the tested format was soluble IgG, and indels increase implementation complexity. Use the residue-preservation insight, not the measured figures as promises for our scFv [R8].
- **Framework engineering, 2025:** A trastuzumab case study found that some distal framework mutations can stabilize antibodies while preserving functionality; generic antibody language-model scores may miss framework effects [R9]. Consequently framework preservation/stabilization deserves a small but distinct portfolio branch.
- **PDB 1N8Z:** Experimental structure contains antibody Fab **light chain A**, **heavy chain B**, and **HER2 chain C** [R10]. These must be mapped to the correct variable-domain ranges, excluding constant-domain residues.

Mutation-numbering warning: published positions such as W99/G103/F104/Y105 and the WGGDGFYAMD H3 segment are described under differing conventions in various studies. Never transfer residue numbers across IMGT/Kabat/Chothia/PDB/scFv indices without an explicit sequence alignment and a mapping table. For core hotspot residues, confirm identities and coordinates before any freezing or substitution decision.

2. Interpretation of the four organizer-recommended resources

Resource	What to use	What *not* to infer
ProteinBase Claude skills [R1]	pdb, uniprot, protein-qc, ipsae, boltz, esm, protein-design-workflow as procedural references and reproducible checks.	A skill page is instructions **not a new predictive model**, a calibrated affinity estimator, or guaranteed working CLI for the user's specific local versions.
Adaptyv Optimization 101 [R2]	Fixed-model versus adaptive model-based optimization; utility of a surrogate; exploration/exploitation rationale; sample efficiency.	Without in-campaign experimental feedback, a conventional Bayesian optimization loop cannot obtain new true K_D data in five hours. Do not call iteration over ESM/Boltz proxies *experimentally supervised* Bayesian optimization.
Adaptyv Optimization 102 [R3]	Experimental expression failure modes; practical reasons for diversification; warning that iPAE is a weak affinity proxy; solubility screening.	Earlier EGFR de-novo-binder hit rates and numeric thresholds **do not directly transfer** to a preexisting sub-pM trastuzumab-derived antibody.
Anthropic Claude binder-design release [R4, R5]	Strong **agent workflow patterns**: immutable inputs, independent review, provenance, model agreement, multi-objective filtering, sensorgram-aware reporting, experiment-first reasoning.	This is mostly **de novo 50–120-residue miniproteins** with **24 hours and up to 2,500 H100 hours** for single targets, not a fair prediction of 5-hour single-24-GB scFv throughput. Its EGFR single-target prompt is a **template**, not a directly valid HER2 scFv

Resource	What to use	What *not* to infer
		experiment contract.

Recommended Claude skills: use pdb, uniprot, protein-qc, ipsae, boltz only if working locally and fast to enable; optionally esm and protein-design-workflow. Claude Code marketplace commands are documented at ProteinBase. Confirm versions and command flags locally. **Skip:** RFdiffusion, BoltzGen, BindCraft and full Mosaic workflows—the challenge is *local editing of a known antibody*, not designing an entirely new binder; setup and model runtime would dominate five hours.

QC-threshold caution: ProteinBase's generic binders' thresholds are useful checks, but the site's example ESM2 PLL > 0 is not meaningful if PLL is expressed as an unnormalized log probability (which is usually negative). Work with the **same scoring definition**, compare to parent and the candidate distribution, and do not apply arbitrary absolute cutoffs from a different model implementation. The same holds for ipSAE thresholds optimized for de novo miniproteins [R11].

3. Existing GitHub pipeline audit and necessary changes

Source: https://github.com/keyshavmor/Catalyst2026Hackathon — read README.md, top_level_orchestrator.sh, Ablang2/.../ablang2_design_config.yaml, ESM/config/esm_pipeline_config.yaml, and Boltz-2/README.md.

Existing implementation	Found behavior	Adaptation or risk
Ablang2/ run_ablang2_mutation_search.py	Large antibody-specific candidate generation; constraints; contextual masked scoring; diversification.	Replace entire parent/position/ref set. Limit generation to **10k–30k** and contextual rescore to a small budget. Previous config has 1,000,000 candidates and a **180-minute AbLang2 runtime target**; inappropriate as-is.
ESM/ run_esm_filter_pipeline.py + multi-stage scripts	Three-stage ESM2-3B scoring; one stage may use FP32; multiple score definitions.	Budget **100–300** candidates, not 5,000. Validate any FP16/BF16 score differences against a small FP32 control subset; no silently substituted model backend. Avoid repeated masked passes for every mutation in thousands of sequences.
Boltz-2/ run_boltz2_pipeline.py	Cofold target + binder, calculate interface metrics; baseline comparisons; fallback MSA modes.	**Force HER2-domain-IV-only cofolding; never full-ECD.** Require full PAE and structure artifacts, independently compute **real ipSAE**, benchmark parent, dynamically scale candidate budget.
top_level_orchestrator.sh	1M → 5k → 300 → 100 structures → 50; hardwired EGFR artifacts.	**Do not run unchanged.** Separate HER2 configuration and stage-local timeouts;

Existing implementation	Found behavior	Adaptation or risk
		avoid hard-coded 241 residue checks and Cetuximab-specific reference distances.
MSA mode	Existing Boltz config calls remote MSA server and warns of rate-limit idle waits.	Precompute/cache target MSA if possible; ensure repeated candidate predictions do not trigger redundant HER2 queries. If MSA is inaccessible, use a documented fallback; validate the parent under that same fallback.

Fail-closed checks: grep/review for EGFR, cetuximab, 1YY9, 241, previous parent sequences, reference distance filters, all 241-position indexing, 50-as-submit rule, and any 1YY9-derived target cropping. These strings may remain in documentation but must **not** affect active HER2 run paths. Do not infer that text search alone proves the code is scientifically adapted—test real input/output on small runs.

4. ML formulation: constrained robust portfolio optimization

Let s_0 be the exact challenge parent and s a candidate. The (unobserved) true improvement is

$$y(s) = \log(K_D(s_0) / K_D(s))$$

which is positive only if measured affinity is stronger (lower K_D). We **cannot reliably evaluate y(s)** without experiment. Instead, use a vector of tractable but imperfect features:

f(s) = [ΔAbLang2, ΔESM2-PLL, target-specific H3 prior,
geometry/contact retention, ΔipSAE or Δinterface PAE,
liability penalty, folding plausibility, sequence novelty].

Use **hard constraints** to ensure the challenge's valid sequence topology, then use conservative Pareto domination, parent-relative evidence, model disagreement and **portfolio diversity**. A scalar score may be produced for triage, but do not conflate its units with experimental affinity; all components require within-model normalization. Prefer rank aggregation over uncalibrated raw sums.

For selecting a set S of 50 from filtered candidates P, a practical heuristic objective is

$$\text{maximize}_S \Sigma_{s\in S} \text{quality_rank}(s) + \lambda\cdot\text{coverage}(S) - \mu\cdot\text{redundancy}(S)$$
subject to $|S|=50$, $s\in P$, branch quotas, and challenge hard constraints.

Where quality_rank is based on Pareto class/robust ranks, coverage rewards distinct mutation mechanisms, and redundancy penalizes identical or nearly identical mutation signatures. This is **not** calibrated Bayesian optimization. A supervised HER2-affinity predictor is optional only if labeled, compatible data are already locally available and can be tested without leakage; otherwise it is scientifically weaker to add one under time pressure.

4.1 Why not train a new model?

The wet-lab signal arrives after the competition, not during the five-hour execution. Training a GP/EVOLVEpro or CNN on binary display labels might be useful if a cleaned dataset and training script already exist, but would transfer imperfectly to this exact scFv format and sub-pM BLI

setting. A realistic 5-hour project should prioritize **evidence quality and throughput**, not an unvalidated model that optimizes the wrong target property.

4.2 Conservative mutation policy

- Initially use **5–8 substitution mutations** per sequence; lengths remain exactly **242**. This upper limit is self-imposed, not an organizer rule.
- Do not mutate the 15-aa $(GGGGS)_3$ linker. Preserve each V_H and V_L domain's correct boundaries.
- Preserve **native cysteines 22, 96, 158, 223** and their disulfides; **forbid new cysteines** in the default run.
- Forbid new N-X-S/T sequons (X ≠ P), screen deamidation, isomerization, unpaired hydrophobics, polybasic patches, extreme charge shifts, and likely buried polar substitutions. **Compare newly introduced** liabilities against parent; do not reject all sequences because the parent contains native motifs.
- Prioritize mutation sites from structure, CDR/IMGT mapping, literature, and AbLang2 proposals. Freeze/highly penalize experimentally established H3 anchors unless targeted evidence justifies a change, with confirmed numbering.
- Beware that "adding harmless framework mutations just to reach five" is *not* automatically safe; multiple substitutions can interact epistatically. Every extra substitution needs a plausible rationale and quality gate.

5. Proposed 50-candidate portfolio

Create the 50-slot manifest at the start, with four hypotheses. Slots are **target quotas**, not a promise that each branch will yield the requested number of safe variants; redistribute only if justified and documented.

Branch	Slots	Mutation hypothesis	Typical composition	Rationale
A — H3 literature-guided	**20**	Focused affinity maturation in well-characterized CDR-H3 neighborhood	5–7 valid substitutions, preferably structurally tolerated	Strongest directly relevant experimental evidence; test multiple plausible H3 motifs [R6–R8].
B — interface/adjacent	**12**	Preserve known contact anchors, optimize flanking CDR-H1/H2/H3 or packing	1–3 interface-near + remaining low-risk V_H/V_L changes	Explores mechanisms beyond H3; structurally informed.
C — mixed affinity + stability	**10**	Combine a small number of H3/other-CDR mutations with compatible support/framework changes	1–3 paratope/near-interface + 3–5 framework	Optimize fitness/developability jointly; avoid destabilizing combinations.
D —	**8**	Retain nearly	5–8	Safeguards

Branch	Slots	Mutation hypothesis	Typical composition	Rationale
framework-biased control/exploration		all binding anchors, make tolerated framework substitutions	framework/adjacent changes	parental geometry and investigates stability/context benefits [R9].
Total	**50**			

Nominated top 15: initialize branch quotas **A:6, B:4, C:3, D:2**, but allow merit-based adjustments after screening. No branch is automatically entitled to a lab slot if it fails structural/quality checks. Use a mixture of very high-scoring and independent hypotheses, not 15 nearly identical neighbors. Aim for different mutated-position sets, amino-acid chemistry and mutation counts. A family with 5 shared substitutions plus one altered residue does not provide 15 independent hypotheses.

Design control: The unchanged parent must be in the *analysis* tables as a reference, **not among the 50 mutated sequences** because it violates the ≥5-mutation rule. Always score parent with the same cropped HER2 target to provide a comparable baseline. The internal 50 is distinct from the ipSAE-complete submission set.

6. Structural preparation, with unambiguous mapping

Primary reference: 1N8Z includes HER2 chain C and antibody Fab chains B (heavy) and A (light) [R10]. Plan:

1. Use the **verbatim organizer CSV sequence supplied above**; write it to parent_scfv.fasta. Verify 242 aa, V_H 1–120, linker 121–135, V_L 136–242, cysteines 22/96/158/223, and the SHA-256 hash.
2. Fetch and checksum PDB 1N8Z and UniProt P04626; strip irrelevant heteroatoms for sequence mapping without losing native glycan/context annotations in the metadata.
3. Extract **variable portions** of Fab heavy and light chains; align separately to the corresponding parent domains. Do not concatenate entire crystallographic Fab chains (constant domains would corrupt numbering).
4. Generate residue_map.csv: scfv_pos_1based, domain, domain_pos, pdb_chain, pdb_resnum, pdb_icode, parent_aa, pdb_aa, imgt_number, cdr_name, min_target_distance_A, contact_class, resolved_in_pdb.
5. Compute distance maps between heavy atoms of HER2 and antibody variable domains; identify direct contacts using e.g. **4.5–5.0 Å**, near contacts **5–8 Å**, and buried core separately. These are *exploratory geometry bins*, not learned affinity thresholds.
6. Extract **HER2 domain IV only** from the PDB/UniProt mapping, including the experimentally observed epitope and enough domain-IV structural context to avoid cutting the fold. **Do not cofold the full HER2 ectodomain**. Specify exact domain-IV boundaries only after mapping; document any terminal or glycan caveats.
7. Compare the parent modeled scFv–HER2 to the experimental complex: epitope-contact recovery, orientation, framework stability, steric clashes, and interface-specific confidence. Do not rank candidate scores before the parent passes a minimal geometry sanity check.

Domain-IV caveat: Cropping accelerates inference but removes receptor context, creates artificial termini, and may omit glycans. Use the **same domain-IV crop for parent and every candidate**, verify the reference epitope and parental geometry, and record these limitations. If the parent fails geometry QC, note scores as low-confidence; do **not** fall back to prohibited full-ectodomain cofolding.

7. Computational generation and ranking stages

Stage G: structure- and evidence-guided proposal pool

Preferred size: 10,000–30,000 sampled combinatorial proposals; filtering is CPU-heavy and relatively cheap **after** the per-site AbLang2 masked scan. Restrict masked scanning to a verified list of plausible mutable sites and retain only a handful of alternatives per site. Mutate fixed positions, preserve high-value anchor sites and the scFv topology.

Construct the four branches independently, not by randomly labeling one homogeneous pool. Each proposal records origin, per-site support, mutation count, parent distance, liabilities, CDR categories, motif lineage, random seed, and all computational model versions.

Avoid unnecessary 20,000-design contextual AbLang2 rescoring and AbRep embeddings. Reserve expensive scoring for ~300–600 diverse proposals if time permits. If AbLang2 cannot load/validate, use only **explicitly labeled** conservative BLOSUM-like/structural candidate generation as a fallback; do not falsely describe it as an AbLang2 model run.

Stage E: ESM2-3B evidence, not experimental affinity

Score a preselected diverse subset (initial cap **150–300**) with the installed esm2_t36_3B_UR50D backend. First benchmark throughput on 10 sequences; then set a hard stop based on available time. Cheap full-sequence log probabilities are preferred first. For a much smaller set (e.g. 40–100 candidates), optionally perform context-masked per-mutated-residue pseudo-log-likelihood and **parent-relative substitutions at identical positions**, in identical precision and preprocessing conditions.

Model memory: 3B parameters alone require roughly 6 GB in FP16/BF16 or 12 GB in FP32, **excluding** activations, PyTorch buffers and caching. A 24 GB GPU may manage one model, but do not overlap a large ESM model and Boltz-2. Prefer mixed precision if model/hardware support it; verify a small subset for numeric stability and no NaN/Inf; do not silently change the foundation model to fit memory.

Select **50 provisional** candidates at this stage, then increase to ~60–80 if there is time and additional structural scoring capacity. Persist a valid final_50_provisional.fasta before allocating the structural budget.

Stage B: Boltz-2 matched baseline and bounded validation

First run the **parent against HER2 domain IV** and measure cold-start latency, peak VRAM, preprocessing/MSA time, GPU inference and completeness of structure/PAE/ipSAE output. Never run a full 630-residue HER2 cofold during this challenge, even for the parent. Never transfer throughput estimates from another complex size.

Target budget: parent + 50 candidate cofolds if the microbenchmark proves this feasible, with **15 distinct scored candidates as the first priority** and 50–100 *sequence candidates* reserved for later cofolding when resources permit. Do not cofold before sequence screening. Run with a consistent domain-IV crop, MSA/preprocessing mode, seed/sampling policy, recycling/precision and metric settings. After parent and first 3 candidates, estimate how many additional complete predictions (including ipSAE postprocessing) fit into the remaining wall-clock. If 50 will not fit, prioritize a diversity-aware 15; never fill remaining ipSAE cells with estimates.

Compute **actual external ipSAE for each candidate intended for submission** from the matching full PAE + structure output, using the verified DunbrackLab/IPSAE implementation (R16), preserving version, numerical cutoffs, score aggregation, chain mapping and residue mask. Do not substitute protein–ligand Boltz-2 affinity predictions: **Boltz-2's small-molecule affinity head is not a calibrated antibody–HER2 K_D estimator** [R12]. Normalize structural metrics

relative to parent. Also inspect contact recovery and new interface clashes/unsatisfied polar groups when practical.

MSA: Reuse target MSA across variants only if the installed software and file format permit. Existing repository warns --use_msa_server can rate-limit and idle the GPU; plan to cache or precompute. An empty-MSA fallback is allowed only as a **separately documented, matched parent-and-candidate protocol** (potentially lower confidence). Do not merge rankings from different MSA conditions uncritically.

Stage Q: multi-objective, diversity-aware portfolio selection

Mandatory first-pass filters: compliant format/sequence, no new cysteines, no linker changes, clear domain mapping, no newly introduced severe liabilities, deduplication, mutation-count check. Second-pass: veto gross interface loss, severe model confidence deterioration, poor structural geometry, extreme ESM2 penalties, or very close duplication. Remaining candidates: assign ordinal ranks or percentile-normalized features *within the same pipeline conditions*.

Example auditable **triage** score, not an affinity equation:

```
rank_score = 0.25 * H3_evidence_rank
           + 0.20 * AbLang2_rank
           + 0.15 * ESM2_delta_rank
           + 0.20 * geometry_parent_retention_rank
           + 0.10 * structural_confidence_rank
           + 0.10 * developability_rank
```

This is a starting hypothesis, not a validated set of weights. If a candidate lacks Boltz results, mark its structural confidence as NOT_EVALUATED and **exclude that sequence from the ipSAE-complete organizer submission**. Keep it in the clearly separate internal 50-row proposal table. Handle screened/unscreened candidates in explicit evidence tiers. Do **not** use branch labels as affinity evidence, and do not select pure geometric winners just because high ipSAE is a judging criterion. For 50, enforce branch-diversity quotas, minimum mutation-set distance where feasible, and limited near-duplicates. For the top 15, require detailed per-candidate justifications and uncertainty flags.

Mandatory ipSAE acceptance gate (not an optional enhancement)

For each scored submission row, require a chain-resolved complex output, full 2-D PAE matrix, and successful external ipSAE calculation. Save boltz_run_dir, structure_path, pae_path, ipsae_stdout_path, ipsae_script_version, pae_cutoff_A, distance_cutoff_A, ipsae_min, ipsae_max_or_directional_if_reported, model_seed, target_domain_iv_definition, and status=PASSED. Each calculated ipSAE must be numeric/finite; reject a candidate if the metric output is missing, if chain indices are wrong, or if PAE/structure residue counts mismatch. Compare a parent baseline under exactly the same configuration and retain raw logs.

Important: Boltz outputs are not affinity estimates. A difference of 0.01 in ipSAE is not a demonstrated K_D gain. The final 15 should balance robust interface preservation, plausible antibody sequence changes, expression risk and mutation diversity; use ipSAE primarily to identify incorrect/broken models and to meet the scoring requirement. If a method requires scores for all 50, budget the number of final sequence proposals dynamically to the number of Boltz runs that can actually complete, rather than submitting 50 unscored sequences.

8. Five-hour 24 GB execution plan and stop-loss decisions

This is a **budget plan**, not a runtime guarantee; all stage caps should be recalculated from local benchmarks. Assume installed model weights and a working GPU as much as possible; if not, setup takes precedence and structural budget shrinks.

Clock	Planned work	Deliverable / decision gate
00–20 min	Freeze exact parent from supplied CSV; map PDB 1N8Z chain A/B/C and **HER2 domain IV only**; check models, weights, environment, GPU.	**Gate A:** 242-aa validated, native cysteines checked, domain-IV crop mapped, organizer requirements frozen.
20–50 min	Port configs, quick AbLang2/ESM smoke test, single **domain-IV** Boltz parent + ipSAE baseline if working.	**Gate B:** observed memory, model load, MSA/structure/PAE/ipSAE throughput; identify fatal blockers early.
50–100 min	Guided AbLang2 mutation scan; diverse 5–8-substitution proposals; sequence QC and independent validator.	**Gate C:** at least 50 distinct valid internal candidates persisted (ipSAE null).
100–135 min	Bounded ESM2-3B scoring; choose 50 ranked proposals and a diverse **15-score-first queue**.	**Gate D:** freeze internal 50 and all sequence-only evidence.
135–265 min	Boltz-2 **domain IV** cofolds, real full PAE + structure + external ipSAE; **finish 15 first**, then expand toward 50 if throughput allows.	**Gate E:** score-complete subset size determined by measured runtime; stop/recover on OOM or queue failures.
265–285 min	Rank only score-complete designs, select top 15 if feasible; independent mutation audit; compose methodology narrative.	**Gate F:** submission_scored.csv contains no missing or fabricated ipSAE values.
285–300 min	Validate output files, scripts, logs, checksums, cutoffs; package and submit honest scored set.	**Gate G:** evidence-backed submission and separate unscored 50 pool.

Hard stop-loss plan: At T+100, preserve 50 valid internal sequences; at T+135, freeze the first 15 score-priority candidates. Never postpone all Boltz work until the end: the organizer **requires ipSAE for submitted sequences**. At T+265, stop GPU inference and export only score-complete candidates. Reserve 15 minutes for compliance. If 15 actual ipSAE values are not feasible, accurately report the smaller scored count and contact organizers early about provided Modal support; never submit invented results.

8.1 Runtime budgeting formula and logging

After timing t_parent and t_variant wall-clock durations (including preprocessing and inference), schedule at most

N_fit = max(0, floor((remaining_structure_minutes - safety_buffer_minutes) / t_variant))

with a dynamic cap (target 50, minimum-priority 15). Re-estimate after each small batch. Log GPU memory.used, memory.total, GPU util, CPU RAM, sample length, PAE output size, model/weights hash, per-seed time, and any MSA waiting. A single 24 GB device should run **one major GPU model at a time**. Delete released model objects and verify VRAM is available before switching from ESM2 to Boltz-2.

Pre-download all required weights **before the contest clock if rules allow**. Also prepare a single conda environment or verified environments, ensure disk capacity for large model caches, and smoke test exact CLI flags. If pre-staging is prohibited, reflect download latency in the 300-minute budget.

8.2 Adaptive fallback matrix

Trigger	Action	How to report
Missing parent official FASTA	Stop; do not fabricate from PDB	BLOCKED_PARENT_SEQUENCE
AbLang2 unavailable	Conservative structure-based/BLOSUM proposals, *not mislabeled* as model-generated	GENERATOR_FALLBACK
ESM2 OOM	Reduce batch size first; use verified half precision; reduce sample count	Preserve model identity and numeric checks
ESM2 too slow	Stop after bounded subset; rerank with available AbLang2 data	ESM_NOT_EVALUATED on others
Domain-IV Boltz OOM	Reduce samples/precision or documented model flags without changing biology; if still OOM, contact organizers about offered Modal assistance	STRUCTURE_UNAVAILABLE, never fabricate ipSAE
Boltz MSA rate limits	Cache MSA or use matched empty-MSA experiment for parent+variants	MSA_MODE=empty flagged
No structural differentiation vs parent	Stop chasing tiny ipSAE deltas; emphasize orthogonal evidence/diversity	No K_D claim
Fewer than 50 valid internal proposals	Broaden conservative mutable positions cautiously; rerun audit	Report shortfall; never fabricate candidates
Fewer than 15 genuine ipSAE outputs	Export only measured score-complete subset; urgently request organizer help if time permits	Never submit null or modeled ipSAE as measured

9. Scientific evaluation, sensitivity and no-leakage policy

Minimum analyses: - Parent vs every candidate: mutation count and site list, domain/CDR mapping, AbLang2/ESM2 delta, liability changes, structural score difference **where measured**, known-contact retention, and uncertainty. - At least one alternative ranking or robustness check:

rank aggregation rather than arbitrary raw-score sums; re-select top 15 with altered weights; report which candidates are stable and which depend on one metric. - Exposure to reference data: if using published DMS, record source and labeling method (binding classification, enrichment, apparent affinity, actual K_D). No disguised single-label conversion to pM units. - Avoid overfitting predicted geometry: compare to known experimental epitope, inspect failure modes, choose some candidates with distinct mutation hypotheses even if slightly lower proxy scores. - Report the 15 candidates' expected risk and evidence rather than asserting any predicted fold improvement in K_D.

Recommended primary scientific claim: "We combined literature-informed, antibody-specific mutation priors with general PLM plausibility, experimentally grounded interface preservation, parent-relative complex confidence, and constrained portfolio optimization."

Claims to prohibit: "Boltz predicts 0.1 pM K_D"; "ipSAE improvement proves higher affinity"; "all 50 are experimentally validated"; "Anthropic's de novo H100 throughput generalizes to a 24 GB RTX"; "dataset binary binding labels equal pM affinity."

10. Output artifact contract for Claude

Directory outputs/HER2_YYYYMMDD_HHMM/:

```
contract.yaml                # rules/parent/source/checksum/compute cutoff
parent_scfv.fasta            # verbatim organizer parent, also hashed
residue_map.csv              # 1-based scFv / IMGT / PDB / CDR mapping
reference_complex.pdb        # source copy, not silent modifications
mutation_site_prior.csv      # AbLang2 + literature + structural priors
all_candidates.csv           # sampled, filtered or eliminated with reasons
internal_50_candidates.csv   # 50 proposals; missing ipSAE allowed here
internal_50_candidates.fasta # 50 unique valid candidate sequences
submission_scored.csv        # only candidates with valid actual ipSAE
submission_scored.fasta      # only candidates in scored CSV
nominated_top_15.csv         # up to 15 actual ipSAE-scored nominations
nominated_top_15.fasta       # score-complete subset of submission_scored.fasta
boltz_parent/                # baseline and all prediction settings
boltz_predictions/           # any completed candidate structures/PAE
model_scores.csv             # per-score definitions, missingness flags
run_manifest.json            # model versions, hashes, seeds, runtime
scientific_readout.md        # explanations, caveats, failure modes
strict_audit.json            # PASS only on complete checks
```

Minimum internal_50_candidates.csv and submission_scored.csv columns:
candidate_id,full_scfv_sequence,sequence_sha256,branch,rank,selected_top15,mutation_count,mutations_1based,mutated_domains,mutated_CDRs,new_cysteine_count,new_NXS_T_count,new_liability_flags,ablang2_score,ablang2_delta,esm2_score,esm2_delta,boltz_evaluated,ipsae_min,ipsae_delta,ipTM,interface_PAE,contact_retention,model_uncertainty,pareto_tier,selection_rationale,source_provenance.

Exact validation tests: 1. Parent equals frozen organizer sequence and has length 242; linker exactly at documented positions. 2. internal_50_candidates.fasta has 50 unique compliant proposals; submission_scored.fasta contains only fully scored variants, each with a finite, independently derived ipSAE and matching PAE/structure artifact links. nominated_top_15.fasta has at most 15 scored selections (aim exactly 15), and all IDs occur in both manifest files. 3. Each candidate is one of 20 canonical amino acids, contains exactly one fixed $(GGGGS)_3$ linker in the

right location, and has valid V_H/V_L boundaries, length ≤260, substitution-only exact length 242 under our initial policy. 4. Each candidate's **substitution count** vs exact parent is ≥5 (default 5–8); mutations reconstructed from manifest give the candidate exactly. 5. Native cysteines preserved, no new cysteines, no newly introduced N-X-S/T sequons; newly introduced serious liabilities vetted. 6. Both domains align to the parent; frozen positions remain unchanged; no duplicate mutation signatures; IDs and CSV/FASTA sequences match exactly. 7. No stale Cetuximab/EGFR input leaked into active files, models or target YAML; files have hashes and provenance. 8. **Every scored submission** has a real structure, full PAE, independently computed ipSAE, metric provenance, PAE/distance cutoffs and a matched domain-IV parent baseline. Internal candidates may have missing scores but can NEVER be labeled submitted. 9. Candidate ordering reproducible under stored random seed and environment; final chosen 15 supported by evidence and limitations. 10. Final sequences can be reconstructed independently by re-applying the recorded mutation list to the exact parent.

11. Claude execution protocol: scoped work packages

Work package 1 — Independent scientific and code audit: Verify this report against the four organizer resources, current publications and the GitHub code; record supported/unsupported claims, alternative approaches, and contradictions. Confirm the official challenge contract and exact parent. Produce audit_findings.md. Do not touch existing EGFR data.

Work package 2 — Target and sequence preparation: Download/verify P04626 and 1N8Z, extract variable Fab segments, map to parent sequence with explicit numbering; identify contacts, disulfides, CDRs, crop and potentially glycosylated sites; generate residue_map.csv and contract.yaml.

Work package 3 — Implement a minimal independent 50-candidate production script: Port AbLang2 generator, apply four branch quotas, hard filters and reproducible deduplication; write provisional 50 **as early as possible**; automatic tests. Do not run 1M/default orchestrator.

Work package 4 — GPU profiling and mandatory ipSAE: Benchmark AbLang2/ESM2-3B; run the Boltz parent against HER2 domain IV and compute real external ipSAE. Then score 15 priority variants before expanding toward 50; persist full PAE + structure artifacts, cutoffs and audit logs.

Work package 5 — Only score-complete submission: Validate parent geometry, score candidates with matched domain-IV Boltz, compute genuine ipSAE for each submitted design, select diverse top 15, and package the design-method narrative and strict audit. Leave any unscored proposals in an explicitly non-submitted internal pool.

Require Claude to emit a timestamped *GO/NO-GO* decision at every stage. When behind schedule, it must **degrade gracefully**, not claim tasks executed or scores measured that are absent. Persist the latest complete 50-person portfolio, all intermediate metrics, and code used.

12. Open questions and falsifiable assumptions for Claude to challenge

1. Does the exact 242-aa organizer parent align residue-for-residue with the variable domains of 1N8Z? If not, which positions differ, and what is the coordinate mapping confidence?
2. Are published trastuzumab CDR-H3 improved variants within five to eight substitutions of this exact parent? Do they involve indels or different antibody format? Can any be used as *sequence priors* without accidental violation of contest restrictions?
3. Which **exact domain-IV residue boundaries** and target-chain sequence are justified by 1N8Z/P04626? Does the domain-IV parent cofold reproduce experimental epitope and orientation? Full-ectodomain cofolding is out of scope.
4. Does ESM2-3B/AbLang2 rank known trastuzumab binders above known nonbinders on a small matched, *non-overlapping* test set? Does adding sequence evidence improve robustness relative to naive BLOSUM proposals?

5. Does Boltz produce reproducible, physically plausible parent geometry? If it cannot, are ipSAE deltas valid evidence at all?
6. Does the 0.82 pM challenge parent use a specific laboratory scFv preparation, orientation and kinetic fit that differs from literature baselines? How should potential assay-floor and rebinding effects be discussed?
7. Are we permitted to preinstall models, download datasets or run preprocessing before the challenge clock starts? If not, budgets must be tightened.
8. Does proteinbase.com/protein-design-skills contain procedures relevant to **local mutation optimization** rather than de novo binder generation? Which installed Claude skills are actually helpful, and which add overhead?
9. What exact file schema does the organizer require for sequence + ipSAE + explanation? The new guidance permits multiple proposed designs and demands ipSAE for **each submitted sequence**, while the earlier slide limits lab testing to 15/team. Submit only real score-complete records.

13. Final acceptance criteria

The challenge run is **successful as a computational workflow** if (i) 50 valid unique *internal* candidate sequences exist, (ii) a scored submission subset has actual ipSAE **for every included design** and ideally includes 15 ranked nominees, (iii) domain-IV mapping and compliance are independently verified, (iv) every reported model score has genuine stored outputs and conditions, and (v) the team explains the method and uncertainty. **No experimental improvement claim** is justified until the assay returns K_D/k_on/k_off and sensorgrams.

References (links and evidence level)

- **[R1]** ProteinBase, *Protein Design Skills* (tooling and procedures): https://proteinbase.com/protein-design-skills
- **[R2]** Adaptyv Bio, *Protein Optimization 101* (model-based optimization, 2024): https://www.adaptyvbio.com/blog/po101
- **[R3]** Adaptyv Bio, *Protein Optimization 102* (experimental competition lessons, 2024): https://www.adaptyvbio.com/blog/po102
- **[R4]** Anthropic, *Claude Protein Binder Design* dataset (de novo binder study): https://huggingface.co/datasets/Anthropic/claude-protein-binder-design
- **[R5]** Anthropic, *Claude accelerates protein design and analytical chemistry* (24 h and H100-hours context): https://www.anthropic.com/research/Claude-accelerates-protein-design
- **[R6]** *AlphaBind, a domain-specific model to predict and optimize antibody–antigen binding affinity*, 2025 (scFv/KinExA data): https://pmc.ncbi.nlm.nih.gov/articles/PMC12296056/
- **[R7]** Mason et al., *Optimization of therapeutic antibodies by predicting antigen specificity from antibody sequence via deep learning*, 2021: https://pubmed.ncbi.nlm.nih.gov/33859386/
- **[R8]** Qiu et al., *Multiproperty optimization of therapeutic antibodies through sequence editing via large language model and reinforcement learning*, Nature Communications (26 Oct 2026): https://www.nature.com/articles/s41467-026-78533-6
- **[R9]** *Tuning antibody stability and function by rational designs of framework mutations*, 2025: https://pubmed.ncbi.nlm.nih.gov/40653772/
- **[R10]** RCSB/NCBI PDB 1N8Z chain/structure reference: https://www.ncbi.nlm.nih.gov/Structure/pdb/1N8Z
- **[R11]** ProteinBase, protein-qc and ipsae procedural skills: https://proteinbase.com/protein-design-skills/protein-qc ; https://proteinbase.com/protein-design-skills/ipsae

- **[R12]** ProteinBase, Boltz-2 limitations/requirements: https://proteinbase.com/protein-design-skills/boltz
- **[R13]** Existing local workflow (GitHub): https://github.com/keyshavmor/Catalyst2026Hackathon
- **[R14]** Anthropic prompt corpus description and its scope: https://huggingface.co/datasets/Anthropic/claude-protein-binder-design/blob/main/prompts/README.md
- **[R15]** Oxford DPhil 2024, *Computational Approaches to Antibody Library Design and Property Prediction* (HER2-aff-large, caveats on transfer): https://ora.ox.ac.uk/objects/uuid%3A463acf7a-feee-4f26-8684-47901764334c/files/d8049g594w
- **[R16]** DunbrackLab, *IPSAE*, authoritative script and implementation details: https://github.com/DunbrackLab/IPSAE
- **[R17]** RCSB PDB 1N8Z, experimental reference: https://www.rcsb.org/structure/1N8Z

Source note: The organizer restrictions in the user-supplied text outrank older recommendations where they conflict. ProteinBase skills are optional execution helpers, not replacements for source validation.

Evidence classification: R6–R9 are primary experimental/model publications; R10 is structural source; R2–R3 are organizer methods blogs; R1/R11/R12 are procedural documentation; R4–R5/R14 describe a larger and substantially different computational campaign. None of the references supplies measured lab performance for the 50 currently hypothetical candidates.